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$$\gamma \sim R[0, 2\theta] \quad \mu\gamma = \frac{0+2\theta}{2} = \frac{2}{2}\theta \quad D\gamma = \frac{(2\theta-0)^2}{12} = \frac{\theta^2}{12}$$

а) Оценка методом моментов: $\mu\gamma = \frac{2}{3}\theta$

Оценка методом макс. правдоподобия: $l(\theta) = \prod_{i=1}^n p(x_i, \theta) = \frac{1}{\theta^n}$

$$x_{\min} \geq 0, \quad x_{\max} \leq 2\theta \Rightarrow \frac{x_{\max}}{2} \leq \theta \leq x_{\min} \\ \Rightarrow \sup l(\theta) = l\left(\frac{x_{\max}}{2}\right) \Rightarrow \hat{\theta}_2 = \frac{x_{\max}}{2}$$

б) Проверка на несмещенность:

$$\mu \hat{\theta}_1 = \mu \left[\frac{2}{3} \frac{1}{n} \sum_{i=1}^n x_i \right] = \frac{2}{3} \frac{1}{n} \mu\gamma = \frac{2}{3} \cdot \frac{2}{3} \theta = \frac{4}{9} \theta \Rightarrow \text{Оценка несмещенная}$$

$$\mu \hat{\theta}_2 = \mu \left[\frac{x_{\max}}{2} \right] = \frac{1}{2} \mu x_{\max} \quad P(x_{\max} \leq y) = P(y) = F(y)$$

$$\varphi(y) = n F^{n-1}(y) p(y) = \frac{n}{\theta} \cdot \left(\frac{y-\theta}{\theta} \right)^{n-2}$$

$$\Rightarrow \mu \hat{\theta}_2 = \frac{1}{2} \int_{-\infty}^{\infty} y \frac{n}{\theta} \left(\frac{y-\theta}{\theta} \right)^{n-2} dy = \frac{n}{2\theta} \int_{\theta}^{2\theta} y (y-\theta)^{n-2} dy = \\ = \frac{1}{2} \frac{n}{\theta} \int_0^{\theta} (t+\theta) t^{n-2} dt = \frac{1}{2} \frac{n}{\theta} \left(\frac{\theta^{n+1}}{n+1} + \frac{\theta^{n+2}}{n} \right) = \frac{1}{2} \frac{\theta n(n+1)}{n(n+1)} = \\ = \frac{\theta}{2} \frac{2n+2}{n+1} \quad \text{Смещение} \Rightarrow \text{Несмещенная}$$

$$\hat{\theta}_2' = \frac{2n+2}{2n+1} \cdot \frac{x_{\max}}{2}$$

Будет несмещенная

Проверка состоятельности

$$D \hat{\theta}_2 = \left(\frac{2}{3} \right)^2 \frac{1}{n^2} n D\gamma = \frac{4}{9n} \frac{\theta^2}{12} = \frac{\theta^2}{27n} \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \text{Оценка состоятельная}$$

$$\mu \hat{\theta}_2' = \frac{1}{4} \mu x_{\max}^2 = \frac{1}{4} \int_{\theta}^{2\theta} y^2 \frac{n}{\theta} \left(\frac{y-\theta}{\theta} \right)^{n-2} dy = \frac{1}{4} \int_0^{\theta} (t+\theta)^2 \frac{n}{\theta} t^{n-2} dt = \\ = \frac{n}{4\theta} \int_0^{\theta} (t^3 + 2\theta t^2 + \theta^2 t^{n-1}) dt = \frac{n}{4\theta} \left(\frac{\theta^{n+2}}{n+2} + \frac{2\theta^{n+2}}{n+1} + \frac{\theta^{n+2}}{n} \right) = \\ = \frac{1}{4} \theta^2 \frac{n(n^2+n+2n+2n+1)}{n(n+1)(n+2)} = \frac{\theta^2}{4} \frac{2n^2+4n+1}{(n+1)(n+2)}$$

$$\Rightarrow D\hat{\theta}_2^2 = \mu \hat{\theta}_2^2 - (\mu \hat{\theta}_2)^2 = \frac{1}{2} \theta^2 \frac{2n^2+4n+1}{(n+1)(n+2)} - \theta^2 \frac{4n^2+4n+1}{4(n+2)^2} =$$

$$= \theta^2 \left(\frac{4n^3+8n^2+9n+1}{4(n+1)^2(n+2)} - \frac{4n^3+4n^2+1}{4(n+2)^2} \right) =$$

$$= \theta^2 \frac{n}{4(n+1)^2(n+2)} \leftarrow D\hat{\theta}_2$$

$$\hat{\theta}_2' = \frac{2n+2}{2n+1} \hat{\theta}_2 \Rightarrow D\hat{\theta}_2' = \left(\frac{2n+2}{2n+1}\right)^2 D\hat{\theta}_2 =$$

$$= \left(\frac{2n+2}{2n+1}\right)^2 \frac{n}{4(n+1)^2(n+2)} \theta^2 \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \hat{\theta}_2' \text{ satisfies the consistency condition and the unbiasedness condition}$$

c) - Проверка эффективности.

$$D\hat{\theta}_2 = \frac{\theta^2}{27n} ; D\hat{\theta}_2' = \left(\frac{2n+2}{2n+1}\right)^2 \frac{n}{4(n+1)^2(n+2)} \theta^2 = \frac{n}{(2n+1)^2(n+2)} \theta^2$$

$$\frac{\theta^2}{27n} \times \frac{n}{(2n+1)^2(n+2)} \theta^2 \Rightarrow \frac{(2n+1)^2(n+2)}{4n^3+12n^2+9n+2} \dots 27n^2$$

$$4n^3 - 15n^2 + 9n + 2 \geq 0 \Rightarrow n \geq 3 \Rightarrow D\hat{\theta}_2' < D\hat{\theta}_2 \Rightarrow$$

$\Rightarrow \hat{\theta}_2'$ более эффективен (при $n \geq 3$)

d) - Проверка гипотезы: $H_0: \theta = 1$ vs $H_1: \theta > 1$

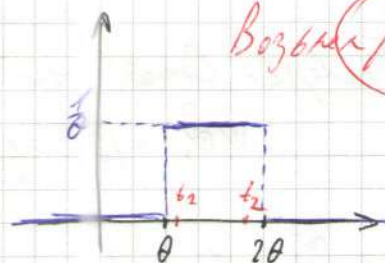
$$t_1 = q_{1-\beta} = \theta + \frac{\theta}{20} \cdot \frac{1}{2} = \frac{41}{40} \theta$$

$$t_2 = q_{\frac{\beta}{2}} = 2\theta - \frac{\theta}{40} = \frac{79}{40} \theta$$

$$F(x) = \frac{x-\theta}{\theta} ; \text{выб } Y = \frac{X_{\max}}{\theta} \Rightarrow Y \in [1, 2]$$

$$\Rightarrow P(Y \leq y) = (y-1)^n \quad y \in [1, 2] \Rightarrow P(t_1 \leq Y \leq t_2) = \beta$$

$$f(\bar{x}_n, \theta) = \frac{X_{\max}}{\theta} = Y_{\max} \sim \varphi(t) \quad \varphi(t) = n(t-1)^{n-1}, \quad f(\bar{x}_n, \theta) = \frac{X_{\max}}{\theta} \sim \varphi(t)$$



Возьмем $\beta = 0.95$

$$t_1 = q_{1-\beta} ; t_2 = q_{\frac{\beta}{2}}$$

$$\Rightarrow P\left(\frac{x_{\max}}{t_2} < \theta < \frac{x_{\max}}{t_1}\right) = \beta$$

$$\int_1^{t_2} (1/t) dt = 0.025 \Rightarrow F(t_2) - F(1) = 0.025$$

$$\Rightarrow t_2 = 1 + (0.025)^{-1/n}$$

$$\text{Итого, найдем } t_2 = (1 + (0.025)^{-1/n})^n$$

⇒ Тотальное гравитационное ускорение: $\left(\frac{x_{\max}}{t_2}, \frac{x_{\max}}{t_1}\right)$

е) Асимптотический гравитационный ускорение при параметре θ

$$\text{Возьмем оценку из ОМТ } \tilde{\theta}_2 = \frac{2n+2}{n+2} \cdot \frac{x_{\max}}{2}$$

$$\Rightarrow \int_{-\infty}^{+\infty} d\left(\frac{\partial F}{\partial \theta}\right) = \int_{-\infty}^{+\infty} d\left(\frac{-\theta - (1-\theta)}{\theta^2}\right) = - \int_{-\infty}^{+\infty} \frac{1}{\theta^2} d\theta + 0 = \frac{\partial}{\partial \theta} \int_{-\infty}^{+\infty} dF$$

Весь все, все, нечего не было результатов → она не была сглажена
сильно результатов → ОМТ не работает

Возьмем $\tilde{\theta}_2$. Св-во: $\tilde{\theta}_2$ — это оценка, которая будет использоваться в ОМТ

$$\frac{\tilde{\theta}_n - \theta_n}{\sqrt{\tilde{\theta}_n - \theta_n^2}} \xrightarrow{d} N(0, 1), \quad \tilde{\theta}_2 = g(\tilde{\theta}_1) = \frac{2}{3} \tilde{\theta}_1$$

$$\Rightarrow \tilde{\theta}_n \xrightarrow{d} \frac{g(\tilde{\theta}_1) - g(\theta_1)}{\sigma(\tilde{\theta}_1)} \sim N(0, 1);$$

$$\sigma^2(\tilde{\theta}_1) = \nabla^T g(\theta_1) K \nabla g(\theta_1) = \frac{4}{9} (\tilde{\theta}_2 - \theta_1^2)$$

$$\Rightarrow \tilde{\theta}_n \xrightarrow{d} \frac{\tilde{\theta}_2 - \theta}{\frac{4}{9} \sqrt{\tilde{\theta}_2 - \theta^2}} \sim N(0, 1) \Rightarrow t_2 = \frac{4}{9} \tilde{\theta}_2, \quad t_2 = \frac{4}{9} \tilde{\theta}_2$$

$$t_2 < \tilde{\theta}_n \frac{\tilde{\theta}_2 - \theta}{\frac{4}{9} \sqrt{\tilde{\theta}_2 - \theta^2}} < t_2 \Rightarrow \tilde{\theta}_2 - \frac{4}{9} t_2 \sqrt{\frac{\tilde{\theta}_2 - \theta^2}{\tilde{\theta}_n}} < \theta < \tilde{\theta}_2 - \frac{4}{9} t_2 \sqrt{\frac{\tilde{\theta}_2 - \theta^2}{\tilde{\theta}_n}}$$

гравитационное ускорение

Получены t_1, g, h сгенерированы в коде.